01 / UNIVERSAL EXTENSION

Personal Earth Engine

GOOGLESTOMI8 / log net D - Telemetry, terrain and personal context

Architecture and reference implementation v0.1
30 October 2026 | Source: supplied log net D master, Edition 2

This addendum turns the requested Earth Engine, POCO/HyperOS, MQTT and Bayer extensions into one explicit system design. The supplied numerical contract stays intact. Environmental rasters, phone measurements and personal history join through time, spatial reference and provenance.

Layer	Outcome in this package
Numerical core	Original machine archive preserved; new compatible coordinate codec and event store tested offline.
Earth layers	Code Editor script for AHN3, Sentinel-2 and Sentinel-1; requires the user's Earth Engine project.
Phone	Kotlin metadata probe for actual sensors, cameras and GNSS capabilities; source scaffold only.
Transport	MQTT 5/TLS adapter and broker examples; no live broker or phone connection established.
Camera	Tested synthetic Bayer interpolation reference; no camera capture or Mali renderer deployed.
Personal data	Per-service integration map; actual Timeline export and selected account access still required.

Decisions that make the design implementable

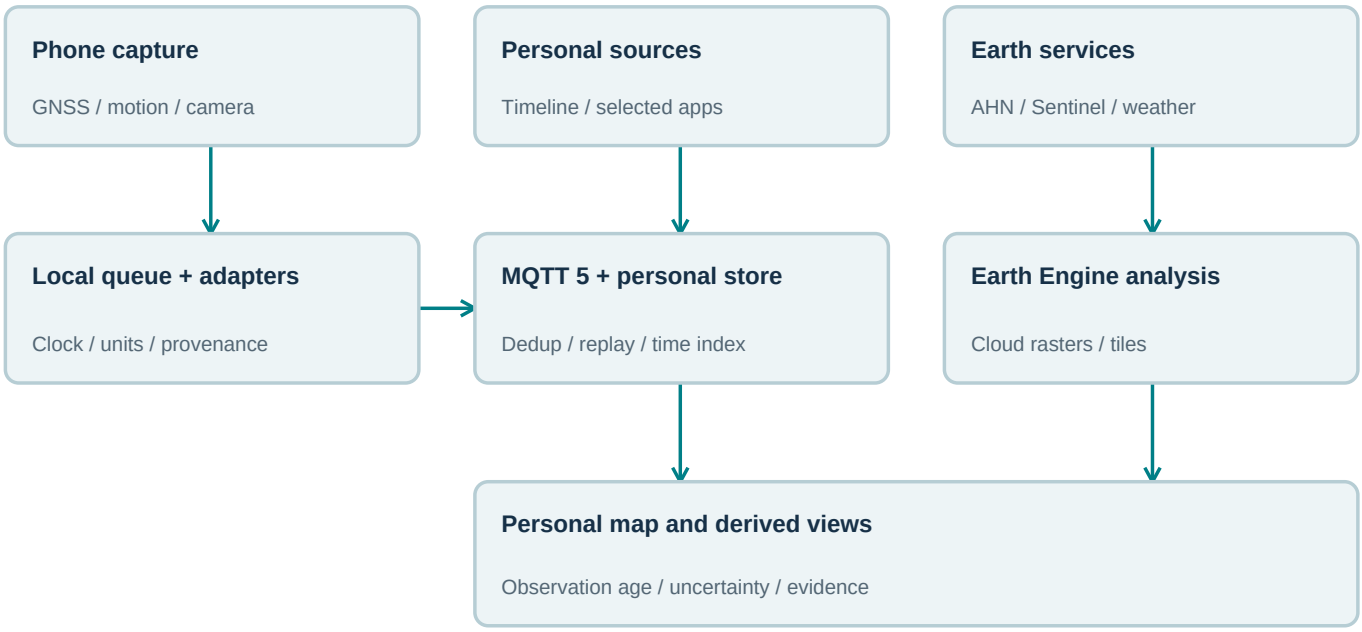
AHN3 is historical airborne laser terrain. Phone GNSS is a live positioning source. Satellite products arrive at provider-specific intervals. These remain separate freshness classes even when all updates enter the same message bus. [S01-S05]

The example handset has LPDDR5X and a Mali-G720. Shared memory still requires cache visibility, synchronization and correct buffer lifetimes. HyperOS retains Android's application isolation and permission model. [S06-S10]

Delivery status: this is a tested local foundation plus integration scaffolding. It is not an installed Android application, a hosted map or an activated telemetry service. No phone, Google account or personal location record was accessed.

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Architecture and responsibilities



The phone owns acquisition, local batching and capture controls. The personal backend owns event persistence, source permissions and private tracks. Earth Engine supplies cloud geospatial calculations and environmental layers. A future map view joins them without turning every motion sample into an Earth Engine asset. [S14]

One event language, several acquisition protocols

Use MQTT for internal publish/subscribe delivery. Provider APIs still use their supported transports: HTTPS, OAuth, webhooks, Google Cloud Pub/Sub and export archives. Source adapters normalize those inputs into versioned events. Satellite data reaches the backend through internet distribution; the phone does not receive Sentinel imagery through its GNSS antenna. [S11,S15-S18,S22-S23]

Bounded local operation

Proposed phone mode: a user-started session, a bounded disk outbox, local feature extraction and explicit stop/pause controls. Network gaps should delay delivery without disguising measurement gaps. Keep high-rate inertial samples local or batch them; publish position and derived features at the rate the view actually needs.

Mapping evidence

A measured fix, a historical Timeline visit, a planned calendar venue and an inferred position are different objects. Store their evidence class. A record with no location remains unlocated. A time correlation alone is not proof that the person was at a place.

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Terrain and satellite layers

Source	Use and freshness	Implementation
AHN/AHN3	Netherlands terrain/surface, 0.5 m pixels; airborne survey period 2014-2019. Static historical reference. [S01]	Included: dtm, dsm and approximate dsm - dtm preview.
Sentinel-2 SR Harmonized	Optical spectral bands; nominal 5-day revisit. Clouds and ingestion affect usable freshness. [S03]	Included: SCL masking, newest candidate and adjustable 90-day median.
Sentinel-1 GRD	Radar backscatter; catalog updates daily and says assets ingest within two days after availability. [S04]	Included: IW, VV/VH, 10 m product filter and one orbit pass.
NASA FIRMS	Near-real-time thermal anomalies; global products generally available within 3 hours of observation. [S05]	Designed adapter; not implemented or connected.
Meteosat FCI	Regional weather context; full-disc scan cadence of 10 minutes. Delivery latency is separate. [S24]	Designed adapter; not implemented or connected.

Vertical reference is mandatory

AHN source products express height relative to NAP. Android Location altitude, when present, is height above the WGS84 reference ellipsoid. Use a suitable geoid/vertical transformation with model identity and uncertainty before comparing them. A generic mean-sea-level value is not automatically NAP. [S02,S25]

```
h = H + N_geoid
height_above_terrain = H_phone_NAP - H_AHN_DTM_NAP
```

Even after datum conversion, the second expression describes the phone antenna's height relative to the historical terrain model; it does not establish a person's height. DSM minus DTM may include buildings, vegetation and other objects and is not a building-only measurement.

Use a small area of interest and zoom-dependent tiles. Preserve source grid and masks for analysis. Do not interpret a 90-day composite as a single new photograph. Keep acquisition, publication, ingestion and display times separate.

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POCO X7 Pro capability model

Treat the handset as the example target until a device report confirms its installed build and public APIs. Official specifications identify Dimensity 8400-Ultra, Mali-G720, 12 GB memory options and LPDDR5X/UFS 4.0. [S06]

Input or output	Expected access and limits
Accelerometer / gyro / compass	SensorManager streams when exposed. Preserve device axes, calibration status, units and timestamps.
Light / proximity / flicker	Hardware is advertised; public stream availability and granularity must be enumerated. Do not assume waveform access.
Gravity / rotation / steps	Virtual or fused outputs depend on the installed sensor list and relevant activity permissions. [S08]
GNSS	Location plus supported satellite-status/raw-measurement APIs. Optional fields vary; receive capability is not a satellite uplink. [S11,S23]
Cameras / microphone	Separate capture APIs and runtime permissions. Camera RAW/CFA access must be queried per exposed camera. [S12-S13]
Wi-Fi / Bluetooth / NFC	Separate APIs and permissions; device/region dependent. These are not arbitrary radio/spectrum receivers.
Touch / battery / thermal	App touch events and available system telemetry; do not equate battery temperature with ambient temperature.
Fingerprint / IR / vibration	Biometrics expose authentication results, not raw fingerprints. IR blaster and haptics are output devices. [S30]
Pressure / Hall	Xiaomi explicitly says this model has no air-pressure sensor and no Hall sensor. [S07]

Capability probe included

The Kotlin source records sensor type, vendor, resolution, range, reporting mode, delay limits and FIFO capacities; camera RAW/CFA/calibration and stream metadata; and GNSS capability flags. It reports missing or permission-limited values explicitly. It does not capture measurements or transmit data.

The probe has not been compiled or run on Android here. A real device report is necessary before choosing sampling rates, RAW processing or optional modules. Installed RAM does not equal available application memory.

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Capture, fusion and timing

Android service model

Continuous acquisition needs a foreground activity or an appropriately declared, user-visible foreground service. Android restricts background sensor access and background starts of foreground services. Camera, microphone and location have additional while-in-use and service-type conditions. HyperOS behavior must be measured on the installed build. [S08-S09]

For Android 12+ targets, certain motion/position listeners are limited to 200 Hz without the high-sampling-rate permission. A permission does not guarantee that the hardware, sensor hub or application will deliver that rate. [S08]

Suggested starting profile	Engineering intent, not measured device performance
Location: about 1 Hz while moving	Reduce rate or use displacement triggers while stationary; keep horizontal/vertical accuracy and provider state.
IMU: 50-100 Hz locally	Batch approximately 100-250 ms or compute local features. Retain sample times within the batch.
Magnetic / orientation: 10-25 Hz	Use calibrated readings, detect disturbances and preserve magnetic versus true-north reference.
Light / battery / thermal: on-change or slow	Avoid unnecessary wakeups; use thermal/power state to adapt rendering and capture.
Camera / audio: explicit sessions	Record only the selected session/mode. Default to metadata/features instead of continuous full-media uploads.

Coordinate frames and uncertainty

Preserve Android device-frame samples, any calibration transform, the local east-north-up frame and the geodetic frame separately. For a true-north clockwise heading, the PDF's east-origin counterclockwise angle is theta = (90 degrees - heading_true) mod 360 degrees. Magnetic heading requires a declared declination correction first. [S26; PDF p6]

Record monotonic event time, a UTC anchor, boot/session identity and clock uncertainty. UTC can jump; monotonic time restarts with the boot. Sensor fusion or dead reckoning creates a derived estimate with lineage and uncertainty, not a replacement raw fix. Extrapolation must expire when its error or age exceeds the chosen application threshold.

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Preserve the numerical contract

The attached master defines a strict representation, not a claim of nanodegree physical measurement accuracy. The embedded archive is retained byte-for-byte. The companion codec implements the same coordinate packing and validates unused codes, domains and padding. [PDF pp3-13]

Original field	Width	Meaning
Latitude / longitude	40 + 40 bits	Signed integer nanodegrees; canonical geodetic domains.
Height	64 bits	Signed millimetres; height kind and datum live in metadata.
Wrapped theta	39 bits	Unsigned local angle; values at or above a full turn are rejected.
Winding	32 bits	Signed turn count; unwrapped phase is derived.
Validity / longitude defined	1 + 1 bits	Aggregate validity and longitude convention.
Padding	7 bits	Zero padding after the 217 data bits: exactly 28 bytes.

```
N = 360000000000 nanodegrees per turn
g(k,b,j) = ((-1)^b * j + k * N/6) mod N
Phi_ticks = winding * N + theta_ticks
```

D6 rotations/reflections are exact on the declared angular or hexagonal structure. Six wrapped 60-degree rotations return the same orientation; six phase advances add a turn. A transformation must declare which operation it performs. [PDF pp6-7]

Measured input boundary

Retain the raw source value and encoding. Exact decimal input can use the original rejection policy for off-grid values. A measured float may be converted only under an explicitly selected quantization rule, with its conversion error recorded separately from sensor uncertainty. The reference supplies rational-error reporting for that boundary.

The frame cannot independently mark missing height, orientation or winding. The new event wrapper adds those availability fields. An absent field uses a zero placeholder, never a measured zero. Wider wrapper quantities use decimal strings in a separately named schema; the original binary profile is unchanged.

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Events and MQTT delivery

The implemented envelope is **personal_earth_event_v1**, carrying one original coordinate frame. It is intentionally narrow; IMU arrays, satellite products, media and app records need their own versioned payload profiles.

Envelope field group	Contract
Identity	source_id, device_id, boot_id and sequence form a stable event key.
Time	UTC with nanosecond text representation, elapsed realtime and clock uncertainty; receipt time is stored separately.
Payload	Original integer_bits_v1 frame as Base64; bounded total payload of 64 KiB.
Availability and frames	Explicit position/height/orientation/winding status, CRS, epoch, vertical datum and orientation reference.
Provenance	demonstration / measured / derived, method, source lineage, raw observation and explicit quantization.
Integrity	Canonical JSON SHA-256 and stored bytes; a digest detects changes but does not authenticate a publisher.

```
lnd/v1/<owner>/<device>/position
```

The supplied adapter uses MQTT 5 over verified TLS, explicit authentication, scoped topics, QoS 1 and non-retained position events. A receiver commits before acknowledging. Same identity and same content is a duplicate; same identity and different content is rejected. [S15-S17]

Offline and stale events

Persist the source event before publishing; allocate sequence numbers durably. Replay after connection loss must preserve identity and capture time. The delivered publisher replays an NDJSON file; it is not a complete Android outbox. Broker acknowledgement is not an application-level proof that a downstream database committed.

Expiry bounds broker queuing; it does not turn old measurements into fresh ones. Mark stale and replayed data at the consumer. Bound queues and report drops. Retained messages may be useful for a short-lived health/status profile, but should not be mistaken for a current private location.

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Google data and the app boundary

There is no universal live feed containing everything in Google or every Android app. Connect each supported source under its own permission and schema, and normalize only the data it actually provides. Android's sandbox applies under HyperOS. [S10]

Source	Supported integration route	Map semantics
Timeline	User-selected device export. Import history after inspecting its actual format. [S18]	Visits/routes and inferred segments remain historical evidence.
Calendar	OAuth, watch notifications and event/delta fetches; renew watches. [S19]	A venue can be a planned place, not proof of presence.
Drive	OAuth, changes notifications and a persisted cursor. [S20]	Files attach to places only when location evidence exists.
Gmail	OAuth, Cloud Pub/Sub notification and history synchronization. [S21]	Selected metadata; an address in text is not a measured fix.
Photos	Picker for selected existing media; Library API for app-created content. [S22]	Use available original metadata and its uncertainty.
Activity history	Takeout or eligible Data Portability archives. Batch/export access. [S27]	Do not present archive arrival as current activity.
Other phone apps	App APIs, content providers, document picker, share/export routes. [S10]	Declare source, rights, time, evidence and missing fields.

Timeline history plus a new live track

Google documents Timeline storage on the device, optional encrypted backups and an Android export path. Use Settings > Location > Location services > Timeline > Export Timeline data. New live tracks should come from the native collector. This design does not assume a public live Timeline streaming API. [S18]

Notification and usage access can expose their respective surfaces; neither grants every app's private database. Keep capture controls, scope selection, retention and deletion in the product. Do not infer a missing location simply to populate the map.

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Bayer processing and Mali rendering

Two independent raster problems

Camera demosaicing reconstructs missing color channels from a declared raw color-filter pattern. Map resampling transforms an existing geospatial raster to a display or analysis grid. AHN and Sentinel products do not become raw Bayer images because they are rendered on the same GPU.

```
RAW + CFA + calibration -> normalized samples -> demosaic
-> linear camera RGB -> color transform -> tone map -> display
```

First query RAW capability, stream format, CFA, black/white levels, calibration and timestamp source for each camera. If RAW is absent, consume ISP-produced YUV/RGB. A marketing binning mode does not establish the raw mosaic layout seen by an application. [S12-S13]

What the included reference does

The Python reference handles RGGB, GRBG, GBRG and BGGR with declared crop origin and row-major black levels. It performs bilinear interpolation using exact rational arithmetic. Synthetic tests check all four patterns and crop parities, known samples, calibration and invalid inputs. It is an algorithm reference, not a full-resolution processor or phone capture path.

Its output is estimated linear camera-channel RGB. Full calibration, separate green-site gains, color-space conversion, demosaic quality improvements, denoising and display tone mapping remain production tasks. Exact arithmetic cannot recover information the sensor never measured.

D6 and square camera samples

A 60-degree rotation generally does not map a square Bayer grid to itself. It needs resampling. Keep the PDF's D6 action for its declared angular/hexagonal domain; label camera interpolation and geometric warps as derived operations. Even a 90-degree raw rotation must update CFA origin/pattern metadata. [PDF p7]

Shared memory still has a synchronization contract

Mali has caches and explicit visibility/order requirements. AHardwareBuffer can share supported memory between components, but buffer ownership, fences, resource layouts and compute-write to texture-read dependencies remain necessary. Profile texture bandwidth, tile behavior, frame time and thermal throttling on the actual phone. [S28-S29]

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What is implemented and verified

Component	Verification performed	Remaining boundary
Original archive	Manifest checksums and sizes; original coordinate fixtures and stored D6/LUT comparisons.	Embedded source code retained as source data; not executed.
Coordinate + event core	Focused offline unit tests: exact packing, domains, quantization, provenance, dedup, collisions and late events.	No device measurement accuracy or network service established.
Synthetic demo	Creates a 28-byte frame and event; stores three events; rejects an identical replay as a duplicate.	Synthetic fixture only; no phone or Google records.
MQTT adapter	Local validation tests and source checks; optional dependency documented.	No broker round-trip, TLS deployment or reconnect endurance test.
Earth Engine script	JavaScript syntax check.	Cloud authentication and server-side execution still required.
Android probe	Public-API source review and capability/version guards.	No Android compilation, APK or actual device run.
Bayer reference	Four synthetic test cases, covering all CFA patterns and crop parities.	No real RAW capture, image-quality calibration or GPU profiling.

Run the local reference

```
python3 -m unittest discover -s reference/tests -v
python3 -m unittest discover -s camera -p 'test_*.py' -v
python3 -m unittest discover -s transport -p 'test_*.py' -v
python3 reference/demo.py --output-dir examples/reference_demo
node --check earth_engine/ahn_satellite.js
```

The code package contains the full numerical archive, new source files, the synthetic outputs, a validation report and this PDF. Core and Bayer checks use the Python standard library. The optional live transport requires paho-mqtt 2.x and your configured broker.

The generic validator cannot prove acquisition or calibration and permits null raw/quantization metadata even for a measured label. Production adapters must require source evidence, conversion policy and structured frame definitions. The reference does not perform geoid conversion or sensor fusion.

SQLite in the reference is a local plaintext demonstration store. Choose encrypted storage, retention and backup behavior before recording private tracks. These are implementation decisions to finish during device/backend integration; the pack does not claim that encryption is already installed.

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Activation sequence and acceptance criteria

Step	Concrete action	Acceptance evidence
1. Environmental view	Register/choose a Cloud project, enable Earth Engine, run the included Code Editor script. [S14]	AHN layers load; source grid and satellite observation times are visible; empty queries handled.
2. Device report	Compile the metadata probe into a native Android app and run on the chosen phone.	Actual sensor/camera/GNSS capabilities and permission-limited results are recorded.
3. One capture path	Implement user-started GNSS collection, bounded disk outbox and versioned coordinate events.	Track timestamps, uncertainty and dropout intervals checked on device.
4. MQTT receiver	Provision TLS identity and ACLs, start a durable broker and the event receiver.	Duplicate/reconnect/replay tests and server commit evidence; no cross-device topic writes.
5. Personal history	Provide a selected Timeline export; implement and verify its schema adapter.	Source counts, times, missing fields and historical/derived classifications reconcile.
6. Additional sources	Add only selected app/API adapters, IMU batches and optional camera sessions.	Each adapter has explicit scope, source lineage and operational limits.
7. Map application	Join private events with environmental tiles; add evidence/age/uncertainty display.	Stale and planned/inferred positions are distinguishable; no invented coordinates.

Inputs that remain external to this conversation

A real device capability report; an Earth Engine Cloud project with access; a private MQTT endpoint and client identity; and a Timeline export or selected per-service authorization. Those inputs determine which adapters can actually be activated. A finished mobile interface and installed background capture service still need implementation.

Production checks tied to concrete failure modes

Test screen-off operation and OS process termination; reboot/clock jumps; network outages and replay; denied/revoked permissions; full storage; bad/out-of-order records; thermal throttling; and vertical-datum conversion. Each resolves a real risk in this architecture. Benchmark on the handset rather than infer throughput from 12 GB RAM or a GPU model name.

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Sources and traceability

Primary references checked on 30 October 2026. Bracketed references in this report identify the sources below.

Attached source: log_net_D_master(1).pdf, 50 pages, Edition 2. Numerical domains: pp3-7; LUT profiles: pp8-10; frame transport: pp11-12; measurement limits: p13. Only this attached PDF was available. The earlier explanatory search PDF mentioned in project history was not available and was not used.

[S01] Google Earth Engine: AHN3 catalog
https://developers.google.com/earth-engine/datasets/catalog/AHN_AHN3

[S02] AHN: Dataroom, coordinate system and NAP
https://www.ahn.nl/dataroom

[S03] Google Earth Engine: Sentinel-2 SR Harmonized
https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

[S04] Google Earth Engine: Sentinel-1 GRD
https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S1_GRD

[S05] NASA FIRMS: global near-real-time availability
https://firms.modaps.eosdis.nasa.gov/

[S06] Xiaomi: POCO X7 Pro specifications
https://www.mi.com/global/product/poco-x7-pro/specs/

[S07] Xiaomi: POCO X7 Pro FAQ
https://www.mi.com/global/support/faq/details/KA-522489/

[S08] Android: Sensors overview and rate/background limits
https://developer.android.com/develop/sensors-and-location/sensors/sensors_overview

[S09] Android: Foreground service start restrictions
https://developer.android.com/develop/background-work/services/fgs/restrictions-bg-start

[S10] Android Open Source Project: Application sandbox
https://source.android.com/docs/security/app-sandbox

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Sources and traceability

Primary references checked on 30 October 2026. Bracketed references in this report identify the sources below.

[S11] Android: Raw GNSS measurements
https://developer.android.com/develop/sensors-and-location/sensors/gnss

[S12] Android: CameraCharacteristics
https://developer.android.com/reference/android/hardware/camera2/CameraCharacteristics

[S13] Android: CameraMetadata RAW capability
https://developer.android.com/reference/android/hardware/camera2/CameraMetadata

[S14] Google Earth Engine: Access requirements
https://developers.google.com/earth-engine/guides/access

[S15] OASIS: MQTT Version 5.0 standard
https://docs.oasis-open.org/mqtt/mqtt/v5.0/os/mqtt-v5.0-os.html

[S16] Eclipse Paho: Python client reference
https://eclipse.dev/paho/files/paho.mqtt.python/html/client.html

[S17] Eclipse Mosquitto: Broker configuration
https://mosquitto.org/man/mosquitto-conf-5.html

[S18] Google Maps: Timeline and Android export
https://support.google.com/maps/answer/6258979?co=GENIE.Platform%3DAndroid&hl=en

[S19] Google Calendar: Push notifications
https://developers.google.com/workspace/calendar/api/guides/push

[S20] Google Drive: Change notifications
https://developers.google.com/workspace/drive/api/guides/push

14 / UNIVERSAL EXTENSION

Sources and traceability

Primary references checked on 30 October 2026. Bracketed references in this report identify the sources below.

[S21] Gmail: Push notifications and Cloud Pub/Sub
https://developers.google.com/workspace/gmail/api/guides/push

[S22] Google Photos: API access changes
https://developers.google.com/photos/support/updates

[S23] GPS.gov: System and one-way signals
https://www.gps.gov/gps

[S24] EUMETSAT: Meteosat Third Generation instruments
https://www.eumetsat.int/meteosat-third-generation-instruments

[S25] Android: Location altitude and timestamps
https://developer.android.com/reference/android/location/Location

[S26] Android: Position sensors and orientation
https://developer.android.com/develop/sensors-and-location/sensors/sensors_position

[S27] Google Data Portability: Time-based access
https://developers.google.com/data-portability/user-guide/time-based

[S28] Arm: Vulkan synchronization on Mali GPUs
https://developer.arm.com/community/arm-community-blogs/b/mobile-graphics-and-gaming-blog/posts/vulkan-synchronization-how-pipeline-barriers-map-to-mali-gpu-hardware

[S29] Android NDK: Native hardware buffer
https://developer.android.com/ndk/reference/group/a-hardware-buffer

[S30] Android Open Source Project: Fingerprint data isolation
https://source.android.com/docs/security/features/authentication/fingerprint-hal

Source archive identity

Embedded archive: log_net_D_bits_bundle.zip
SHA-256: 058c870f9171d12c0d298928126ec28f9c35da643a9fb5c5c7c9e1302d7fca08

The preserved archive retains the original supplied content and synthetic numerical fixtures. New demonstration events do not identify the user. Proposed sampling rates, architecture, topic names and activation steps are engineering choices in this addendum, not measured device performance or provider guarantees.